

Research Report: Data Ownership

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Introduction

Within today's digital world, almost everything we do online produces data. From search history, location, social media, and browsing activity, individuals are constantly producing information that companies collect, store, and monetize (Zuboff, 2019). Despite the personal nature of this data, users currently have little to no legal ownership over it. Companies have found a way to gain value from this data via targeted advertising and by monetizing it, often without meaningful consent or compensation (Zuckerman, 2014). This raises the question of whether individuals should have the legal right to own the data they generate online, even if doing so disrupts the current model of "free" internet services.

The issue of data ownership begins with the inherent imbalance of privacy, economics, and digital rights within the digital realm. Ownership has been tied to creation, where if an individual produces something, they have the power to control it (Jurcys et al., 2021). However, this idea has not been consistently applied within the digital space. As users continue to generate large amounts of personal data through their online presence, corporations claim control over that data once it is collected. This has led many professionals and researchers to deem our current economic system a form of "surveillance capitalism," in which personal information is treated as an important resource for companies to easily be scraped and sold (Zuboff, 2019). As user concerns over the use and control of their online data persist, the debate over the structure and lack of user data control becomes increasingly prominent.

Users should have full ownership over the data they generate online, including the right to decide whether it is to be collected, analyzed, sold, or deleted, even if this heavily changes how web services operate. User data is tied to personal identity and should be treated as a form of personal property. The current system in place creates a significant vulnerability for users, making it nearly impossible to fully protect personal information once widely collected. By creating clear property rights over data, digital economic markets will improve by limiting overcollection from companies and encouraging ethical and transparent data exchange between users and companies. While this shift may fundamentally challenge the foundations of "free"

internet services, it will bring fairness and accountability and ultimately grant individuals full ownership over their data in this digital age.

Literature Review

To start the literature review, it is necessary to establish what user data is. User data is a combination of personal information generated online. The personal information includes information explicitly shared online, like on social media apps, information that is obtained by observing activities of the user, like locational data and browsing history, and information that can be inferred from the user, like credit scores (Fischli, 2022). The amount of data being produced by users has been increasing daily due to the growth of the internet as a whole. Much of this data collection happens on social media apps by tracking users using algorithms and collecting information from the way users interact on the app. A way that data is collected is through data scraping which is an “automated data collection” process that uses software to collect large amounts of data online (Brown et al., 2025). This process is done without the awareness of the user. Once collected, that data is monetized by big companies for profit and allows users to use their platforms (Zuboff, 2019). Monetization can come from data brokers which are companies that prioritize users’ data for money.

Companies rely on user data, and this has resulted in discussions of ownership of user data. One reason is that people want to know who has control over their data whether it’s the users themselves or companies. Researchers are examining the issues and solutions that pertain to user control of data and ownership. By looking at the different perspectives that relate to ownership of user data in the literature review, we can understand how different sides view this issue. The first major issue is whether user data qualifies as personal property. User data being considered personal property is discussed because there is no clear law that states whether user data is personal property or not (Jurcys et al., 2021). As a result, two different perspectives are formed. Perspective one claims that user data can’t be considered personal property, since there is no clear law defining user data falling under personal property, and governments would have difficulties enforcing this. This is due to the government operating on legal definitions that have

been written into law, and there isn't a clear definition under the law for user data (Jurcys et al., 2021). This is how the government goes about user data currently. Another perspective proposed is that user data is personal property because it has "clearly delineable boundaries, (b) provides economic value to the individual, and (c) can freely be disposed of" (Jurcys et al., 2021). Defining user data within these three categories would make user data fall under the category of personal property. Continuing with this proposal, researchers like Fischli also say that specifically in the West, a property-owning democracy is in place, which is one that allows "all citizens in a position" to own property, whether it's "social and economic" (Fischli, 2022). Since a property-owning democracy is being used, and data falls under personal property, users should be able to own their data, according to researchers like Fischli. A property-owning democracy is a system where property and wealth are meant to be distributed among citizens instead of a few massive corporations. If personal data has financial value and functions like property, then users, not corporations, should have ownership rights over it in a property-owning democracy.

Another important idea is that user data holds economic value that is important to big companies. In the US, three out of the five richest men have built their wealth through "data-driven companies and technologies" (Fischli, 2022). To expand their wealth, they need user data for their companies. Companies like Google, Facebook, and Amazon get data by collecting data from users and using that data to profit themselves (Zuboff, 2019). As a result, people have their personal identity taken by these companies and used for monetary gain. This model is referred to as "surveillance capitalism," which emphasizes the need for data for companies (Zuboff, 2019).

Something that was prevalent in the research was that there isn't a lot of user control over personal data, and users don't know the type of data that is being collected from them, raising concerns with the public. Many users don't read the terms of agreement of sites or applications that they are using. A Pew Research study on whether Americans read privacy policies for companies shows that only 9% of Americans always read through them, 13% often read through them, 38% do it sometimes, and 36% never read through them (Auxier et al., 2019). Of the 60% of Americans who read them to some extent, only 22% of them read through them thoroughly,

with 35% reading parts of the policies, and 43% skimming through them without a thorough read (Auxier et al., 2019). Even though users consent to these terms, they skim or even skip over the terms due to their complexity and length. Something to mention is that the reports in the survey are self-reported. This means that the results may reflect self-reporting bias. Also, most people don't know their own records, including "criminal records, fitness data, or credit history," which many people think they have control over (Fischli, 2022). This shows a disconnect between what is legally true and what users truly believe. Personal data consists of many details, which include information shared explicitly, like social media profiles, recorded activities like browsing preferences and location data, and analyzed personal data to infer additional data from individuals, like the calculation of credit scores (Fischli, 2022). Data can be collected even if users never directly provide it themselves.

Companies collect "large amounts of sensitive and personal information," which has resulted in users being left vulnerable (Alessi et al., 2018). An example of a scandal is the Equifax breach of 2017. Hackers were able to get into the agency's internal system and obtain "names, social security numbers, birth dates, addresses, and, in some instances, driver's license numbers" of over 145 million Americans (Kenny, 2018). This leads to the risk of misuse or exposure of private user data. The large-scale collection of data also creates significant risks for users, as when data is misused or exposed, victims' sensitive personal information can be used to target them online and in real life.

Companies aren't perfect, and cyberattacks are bound to happen. Cyberattacks may also result in data being exposed, leading to harassment from others online, specifically doxing and swatting. Doxing is when "someone's personal information is shared on the internet without someone's consent," and swatting is the act of creating a "fictitious report to the police" and having officers come to the victim's home without warning (Li, 2017). Exposed data may include personal information, such as home addresses, phone numbers, or workplace information, which can be used to target victims online or in real life. Since companies collect such large amounts of user data, a single data breach can expose sensitive information belonging

to millions of people. Doxing and swatting are not prosecuted often enough because there are “inadequate laws” and a “shortage of resources to deal with these crimes” (Li, 2017). Because of the lack of laws protecting users from doxing or swatting, victims may not be able to find the culprit and get them arrested. This demonstrates how the lack of control users have over their personal data can create serious safety and privacy risks once this information is online. Also, every user of the internet can be a victim of these attacks because the data is on the internet (Li, 2017).

Possible solutions are being proposed by both researchers and tech companies. Many companies are adopting different methods of encrypting data and protecting privacy. Cloud-based systems use a “public key encryption” (PKE) that only the user of the data can use to decrypt the data (Mehmood et al., 2016). This makes data on the cloud unreadable to unauthorized parties, even if the data is intercepted. Companies are also using “anti-tracking extensions, advertisement/script blockers, and encryption tools” that block access to user data (Mehmood et al., 2016). When it comes to data being stored on personal computers, anti-virus and anti-malware programs are being installed. Although no method guarantees the protection of user data, some can minimize the risks and increase data control for users.

Critics of data ownership argue that granting users full control would disrupt the model of free internet services. These models rely heavily on advertising revenue that is generated from user data (Zuckerman, 2014). Many online platforms, like social media and search engines, run without direct cost to users because companies monetize personal data instead. If users got full ownership, companies would lose access to a lot of data. This may lead to subscription-based models to make up for the financial losses. Most of these companies have already built their entire infrastructure on algorithms that operate on large-scale data collection. Private companies enjoy having power over personal data, so they may not be likely to sign public contracts that allow users to own their data (Fischli, 2022). However, this perspective ignores the fact that these services are not truly free, as users already pay with personal data. Granting ownership to users won’t get rid of digital services but instead would encourage transparent business practices.

Companies could offer subscription-based services or voluntary data-sharing agreements for users who want to give up their data. Moving toward these methods will prioritize user consent while maintaining access to these platforms.

The research shows how complex the issue of users owning their own data is. Users have little control over it compared to the companies that use their data, despite being the primary source. Even with the little control they have, their data can be misused and subjected to cyberattacks that they may not be fully protected from. Companies are incorporating their own methods to keep data private, but these methods do not always guarantee protection. Increasing user control over their data has technical and economic challenges, but it remains an open discussion.

Preliminary Argument

Firstly, we argue that user data is fundamentally personal, and living in a society that supports a property-owning democracy grants users the right to have their data treated as personal property under the law. User data reflects online activity that is “observed, observable, and derived from different services and activities which are stored on an individual’s personal data cloud” (Jurcys et al., 2021). This reflects information that is unique to the individual and cannot be replicated by other individuals or companies. With billions of people being online every day, the amount of data generated is enormous. Companies capitalize on this for profit, leading to a market where our data is free labor for these companies. This leads us to surveillance capitalism, as described by Zuboff (2019), a situation where our human experiences, which include our digital interactions, become raw material for translation into behavioral data. This data, through manufacturing processes, is then converted into prediction products, which allow companies to predict the future behavior of users. For instance, through these prediction products, companies are able to track user activity and use these “insights” to do things like show you personalized ads (Federal Trade Commission, 2023). While users are promised total certainty and are allowed to use the internet freely, we do so at the expense of our democracy, freedom, and our human future (Zuboff, 2019). Should this be the case if we live in a society that

supports a property-owning democracy? The primary goal of property-owning democracy as described by Fischli (2022) is to put “all citizens in a position that affords them the right to manage their own affairs to a suitable degree of social and economic equality”. By granting users ownership of their data and treating the data as property, not only do we allow users to manage their digital lives, including how these companies use their data, but we also strive toward a more equitable society.

The lack of meaningful ownership of user data leaves users vulnerable. Users can’t protect themselves online, especially when companies collect their personal data, whether or not companies say so. Data that has been exposed can lead users to different forms of online harassment. Examples include swatting and doxing. Referring back to the literature review, doxing is when “someone’s personal information is shared on the internet without someone’s consent,” and swatting is the act of creating a “fictitious report to the police” and having officers come to the victim’s home without warning (Li, 2017). Specifically with doxing, victims may not know what data is being exposed. Data that is personal to you could be shared online without your awareness or permission. In addition, the lack of federal laws makes it harder for users to protect their data and find the culprit if they are victims of the attack (Li, 2017). As a result, users are left unprotected, and their data is now exposed if they are victims of these attacks. Most instances of online harassment, if not all, result from the disclosure of personal information (Aliyu et al., 2023). Allowing users to own their data will help people take action against forms of online harassment and attacks to keep their data safe.

Ownership of user data would improve how digital markets operate for users. Users are able to use the internet for free in exchange for their data and for seeing targeted ads (Zuckerman, 2014). Companies today are able to follow this same framework and gather data on millions of users around the world. This causes a large amount of data to be collected, even if it’s unnecessary, inaccurate, or irrelevant. If users owned their data, companies would obtain it through explicit consent. That would naturally reduce overcollection and shift the system towards data that is higher quality and voluntarily provided.

Moving from data collection to the data itself, it is clear that the data is very valuable financially. Studies have found that while it is hard to determine an individual's personal data worth, estimates range from a couple of hundred dollars to thousands of dollars (Bloor, 2020, as cited in Fischli, 2022). Referring back to the literature review, three of the five richest men in the US built their wealth through collecting data for their companies (ATF and IPS, 2021; Collins, 2021, as cited in Fischli, 2022). This shows the financial importance of data because it's worth a lot by itself and helps make the rich richer. As Fischli (2022) describes, while some actors could incentivize this and cause users to share more information about themselves for money, what we propose is a framework that allows users to earn money from data they have consented to share with companies. With this framework in mind, things like targeted advertising would not disappear; it would just become more precise and ethical. Understanding the value of data and its impact on the internet can help companies make new frameworks so that users can have the option to own their data.

Conclusion

The question of data control largely comes down to how we define personal property, how we protect users and their data from vulnerability, and how we improve the digital market. Users online should have the legal right to own the data they produce. Personal data is the reflection of people's identities, actions, and behaviors, and is unique to each person. Allowing companies to freely collect and profit from this data then creates an imbalance that fosters neither privacy nor economic fairness.

By deeming user data a form of personal property, individuals would gain greater control over how their information is used and shared and whether it is monetized. This development would also help address data exploitation by companies, reduce online harassment, and limit the vulnerabilities created by large data collection. Additionally, creating clearly defined ownership rights would promote a more efficiently run digital marketplace that forces companies to rely on transparent data exchanges instead of freely collecting data on their users. Although this change

may disrupt the current model of “free” internet services, it would lead to a system that prioritizes user ownership while holding companies accountable.

This paper focused on establishing why users should legally have full ownership over the data they generate online, even if it affects the current model of the internet. However, there are certainly areas that should be further developed to reinforce the argument of data ownership. Most notably, this includes exploring real-world examples of data ownership in practice and further researching subscription-based models and voluntary services as an alternative to “free” platforms. These ideas are mentioned but not fully explored in this paper, and may prove to be extremely significant as the issue of data ownership continues to advance.

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